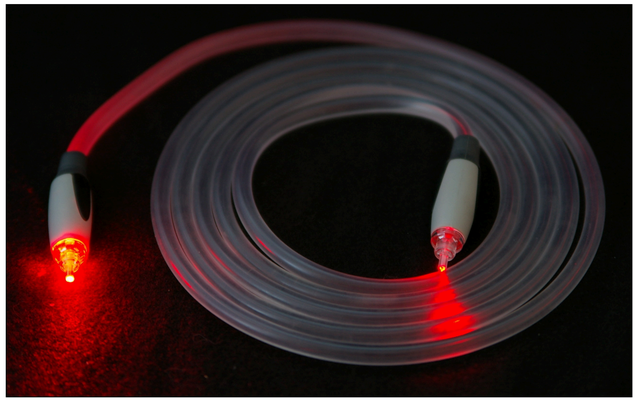
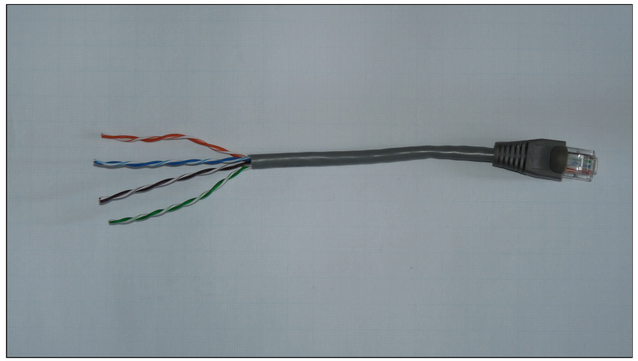
physical

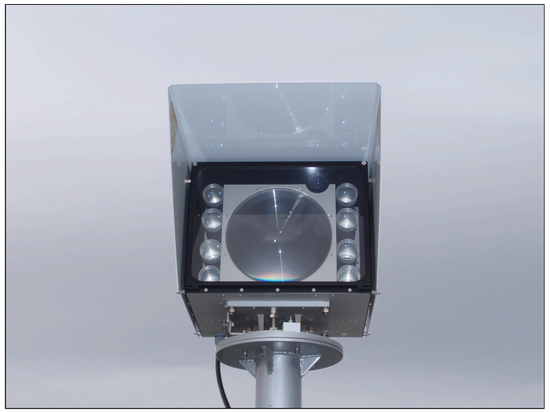
physical media



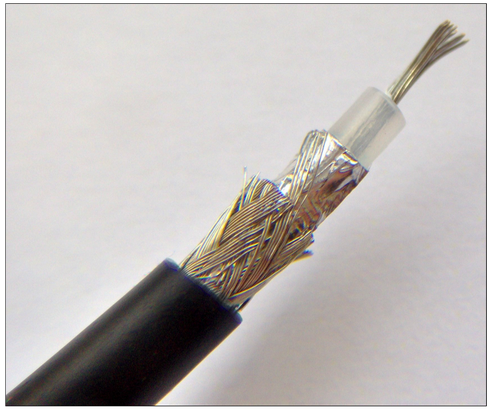
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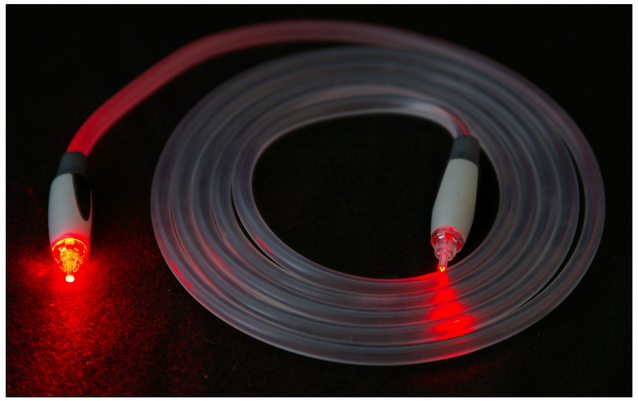


Evan-Amos, via Wikimedia Commons

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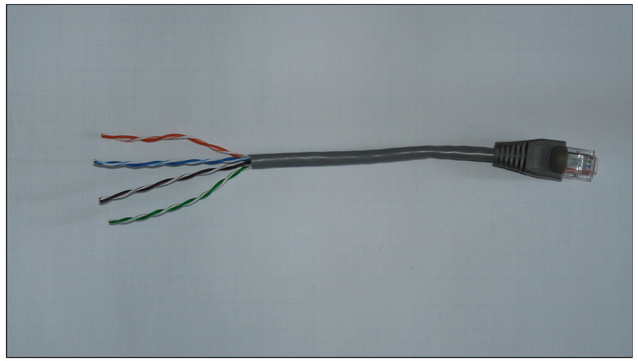
physical media

fiber carrying light



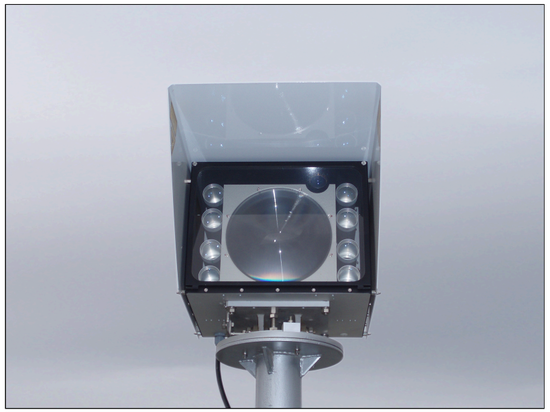
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bundle of wires



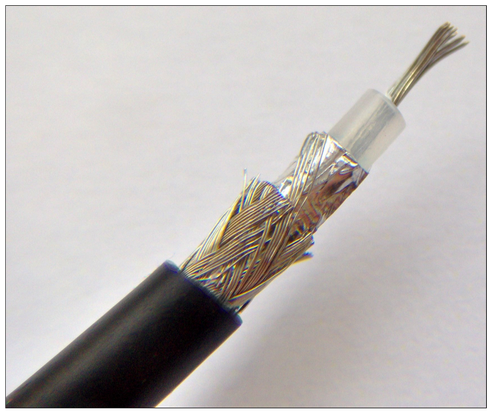
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infrared through air



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single wire



FDominec, CC BY-SA 3.0, via Wikimedia Commons

radio



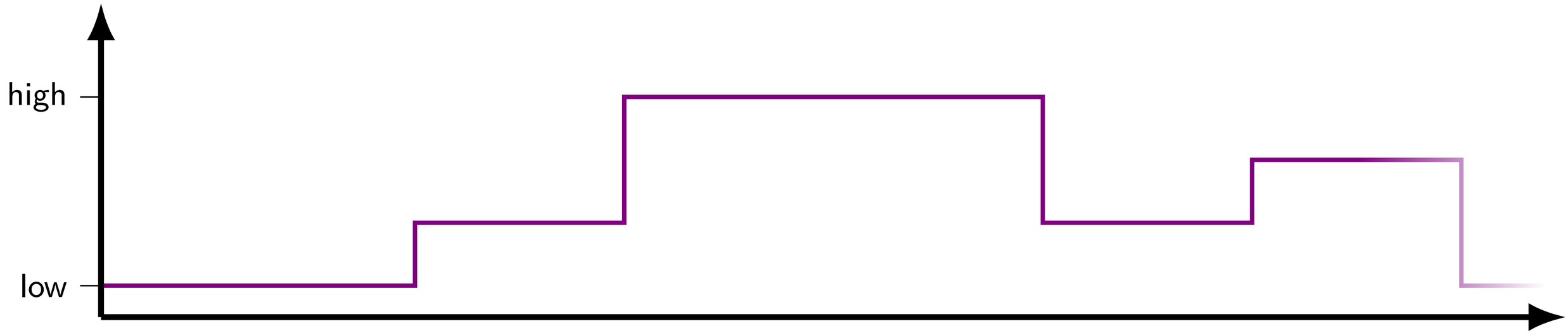
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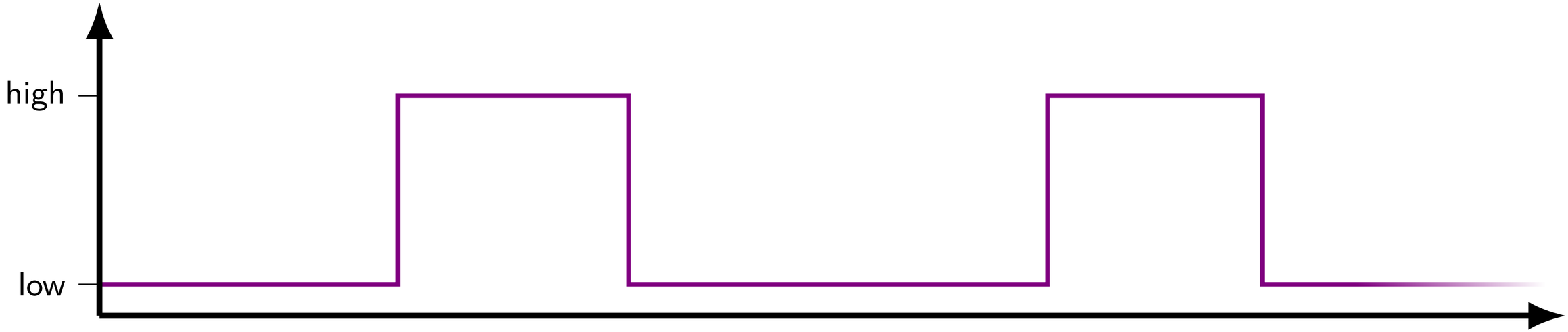
transmitting a signal

can vary…

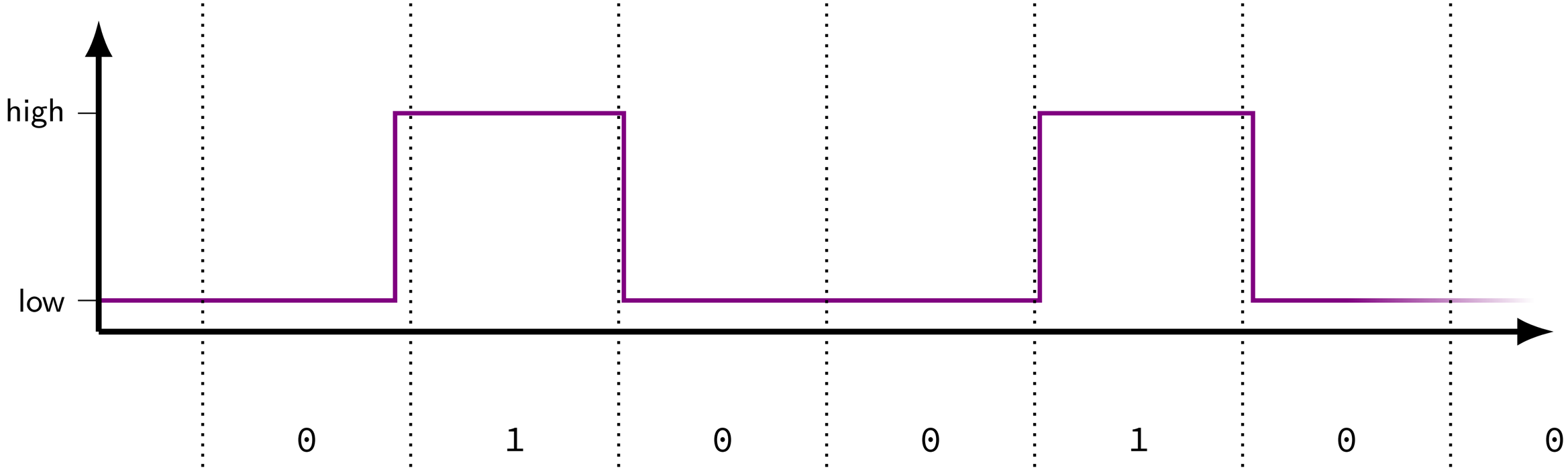
- voltage
- radio/light intensity
- radio/light frequency
- …



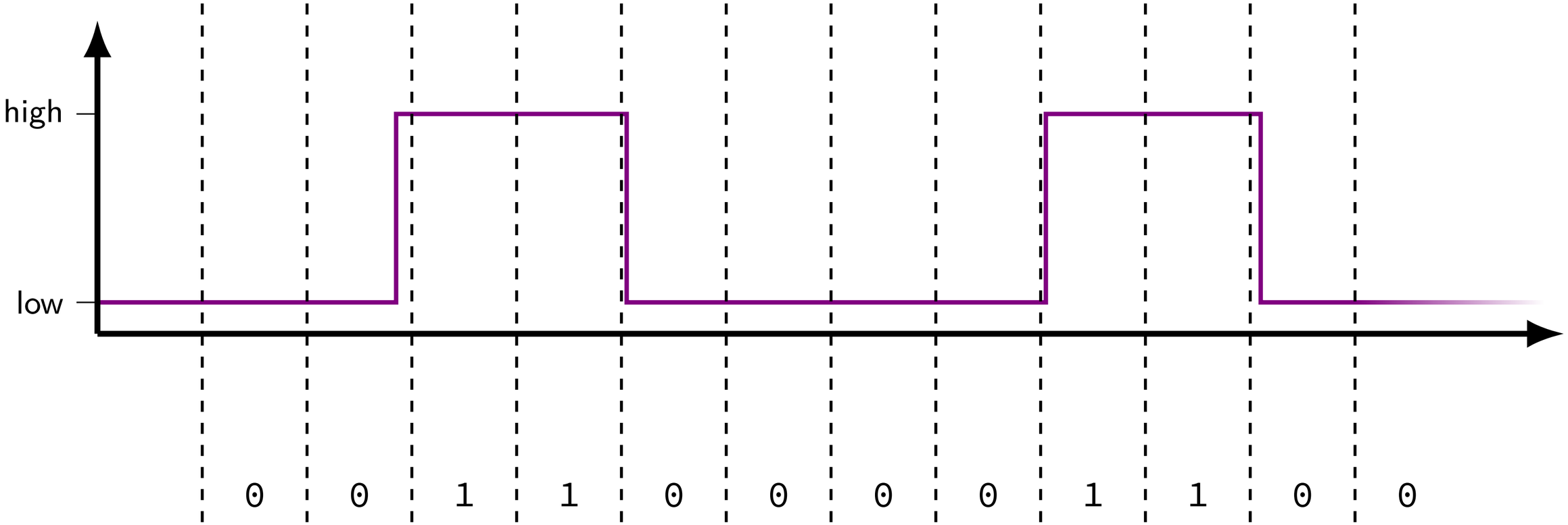
clocking



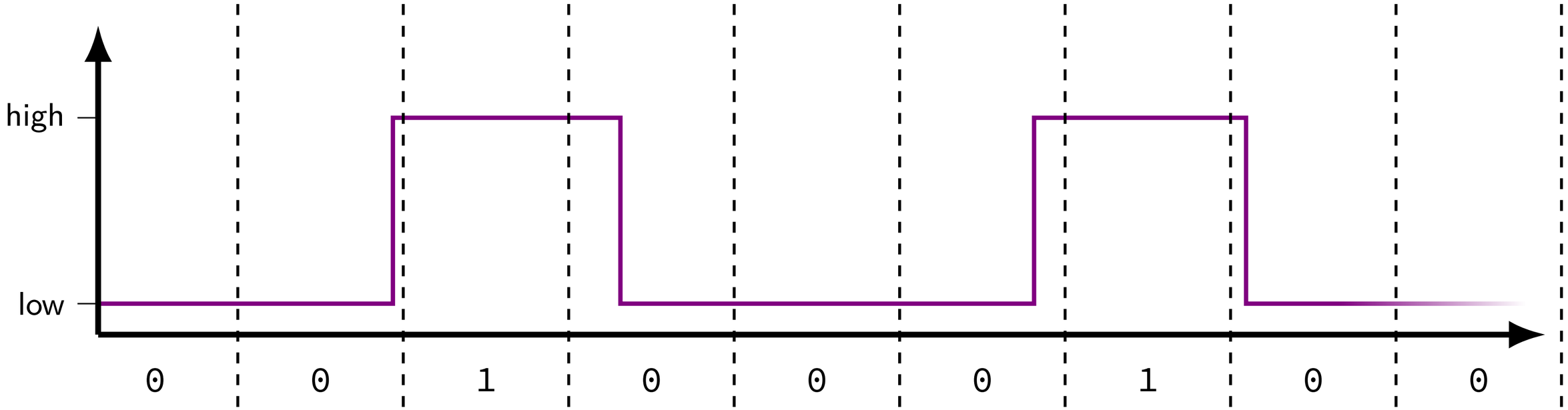
clocking



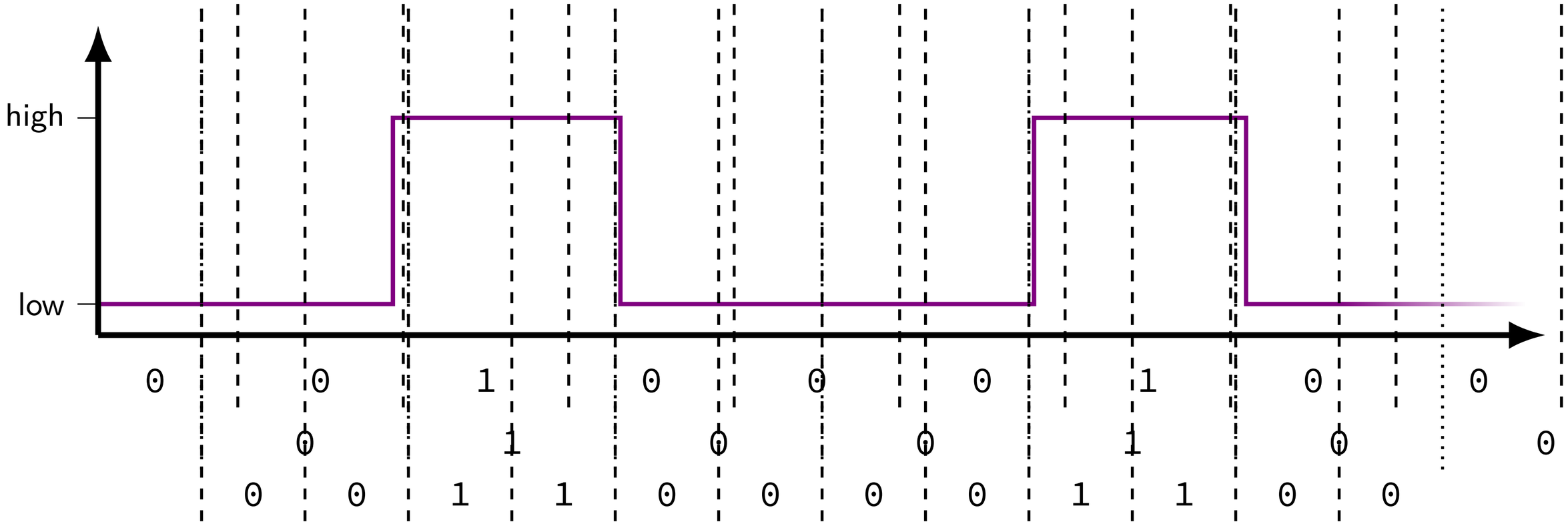
clocking



clocking



clocking



better encodings

need to keep sender/receiver in sync

- may enforce having frequent transitions to resync
- (avoid long strings of all-low/all-high)

transmit more than just 0/1 at each time:

- use more levels than just low/high
- vary more than just one thing
  - example: pulse amplitude and duration
- may allow for “control” symbols

probably covered in ECE Signals course?

Backup slides

simplest case

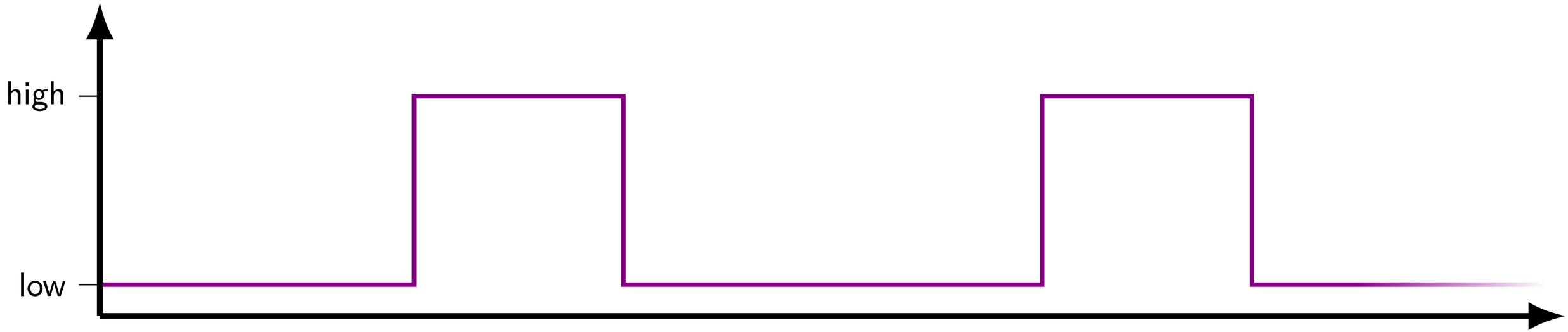
the simplest case

signal low/high — no in between

only one ‘channel’

- won’t have multiple wires/antennas/frequencies/etc.
- won’t modulate different things same time

want to send receive bits (0 or 1)



self-synchronizing?

can resynchronize clock by looking for transitions

but doesn’t work if lots of consecutive symbols

also, need to know where low and high point is

important to have transitions to low/high to calibrate this

self-sync and 'start-message'

one idea: set start-message to have lots of 0/1/0/1/0/1/0/1/etc.

- something 10Mbit Ethernet does

but probably not enough to ensure things stay in sync on big message

alternate strategy — use encoding that adds extra transitions

example: Manchester encoding for 1-bit

Manchester encoding

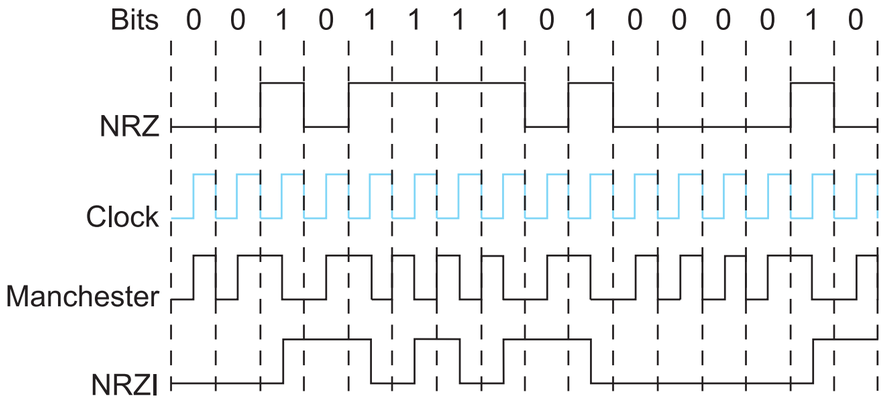


Figure 25 from Section 2.2 of Computer Networks: A Systems Approach (6th ed) (Peterson and Davie)

problem with Manchester

fixed the problem of too few transitions

but now most transitions don't send information

means we aren't making good use of wire/etc. capacity

there are more clever compromises

(example: 4B5B encoding)